Data Quality Control Report

Scope and objectives

This report documents the quality control (QC) procedures applied to the ANDD dataset to ensure consistent annotation, biological validity, and traceability of antibody-derived molecule records. The QC process focused on (i) harmonizing immunoglobulin format annotations using a controlled vocabulary, (ii) auditing cross-source consistency, and (iii) quantifying dataset composition by molecule format and species. A representative example of ANDD is shown in Fig.1.

Source	Update_Date	PDB _ID	PDB_ID_ Changed	Experimental_ Method	Structure_Title	Complex_ Structure	Ab_or_ Nano	Source_ Organism
AB_Bind	2015/10/6	1DQJ	No	X-RAY DIFFRACTION	CRYSTAL STRUCTURE OF THE ANTI-LYSOZYME ANTIBODY HYHEL-63 COMPLEXED WITH HEN EGG WHITE LYSOZYME	TRUE	Antibody	Mus musculus

H_Chain Entity ID	H_Chain Asym ID	H_Chain Auth Asym ID	H_Chain Database Name	H_Chain Accessio Code(s)	H_Chain Sequence Cluster ID	H_Chain Sequence Cluster Identity Threshold	H_Chain Macromolecule Name	L_Chain Entity ID	L_Chain Asym ID	L_Chain Auth Asym ID	L_Chain Database Name	L_Chain Accession Code(s)
2	B	B	UniProt	P01865	16799	100	ANTI-LYSOZYME ANTIBODY HYHEL -63 (HEAVY CHAIN)	1	A	A	UniProt	P01837

L_Chain Sequence Cluster ID	L_Chain Sequence Cluster Identity Threshold	L_Chain Macromolecule Name	Ag_Entity ID	Ag_Asym ID	Ag_Auth Asym ID	Ag_Database Name	Ag_Accession Code(s)	Ag_Name
9340	100	ANTI-LYSOZYME ANTIBODY HYHEL-63 (LIGHT CHAIN)	1	A	R	UniProt	P16218	guanine nucleotide-binding protein subunit alpha-3

Ag_Seq	Ag_Source Organism	Ab/Nano_ Mutation	Ab/Nano H_Chain AA	Ab/Nano L_Chain AA	Ab/Nano_ CDR H1	Ab/Nano_ CDR H2	Ab/Nano_ CDR H3	Ab/Nano_ CDR L1	Ab/Nano_ CDR L2	Ab/Nano_ CDR L3	CDR Nomenclature
MDYKD ... LFKKIS	Acetivibrio thermocellus	no	EVQLQ... VTVSA	DIVMT... KLELK	DYYIH	WIDPE..P KFQG	\	KASQNV GTAVA	SASNRY T	QQYSSY PLT	Chothia

Affinity_K_d (M)	$\Delta G_{binding}$	Affinity_Method	Reason_Code	Predicted_or_ Not	Provenance
1.68735E-14	-18.7781	SPR	not_reported	real	abYbank_ab|PDB|schema_harmonisation

Note: "... " meaning that for the sake of brevity, multiple amino acid sequences are omitted here, but they are fully included in the ANDD dataset.

Fig. 1 **An showcase of ANDD.** *ANDD unifies fragmented sequence, structural, antigen, and affinity data from diverse sources into a single, standardized resource.*

Introduction and Controlled Vocabulary

The ANDD dataset aggregates antibody and antibody-derived molecule records from multiple heterogeneous sources, where immunoglobulin format annotations are often inconsistent or biologically ambiguous. In addition to full-length antibodies, these sources frequently include light-chain dimers (BJ), nanobodies (VHH), and single-chain variable fragments (scFv). In particular, antibody-derived binding formats such as nanobodies (VHH), single-chain variable fragments (scFv), and light-chain dimers (BJ) are often conflated or incompletely described.

To ensure consistent, biologically accurate, and reproducible annotation, we defined a controlled vocabulary for immunoglobulin formats using IMGT as the standard authority.

This vocabulary defines four mutually exclusive categories—H2L2 antibody, scFv (single-chain variable fragments), VHH (nanobody), and BJ (light-chain dimer)—based on chain composition,

domain organization, and biological origin. For each category, formal definitions, required inclusion criteria, and explicit exclusion rules were documented in Table 1 to ensure that annotation decisions are reproducible and interpretable by downstream users.

The controlled vocabulary was applied uniformly across all ANDD records and served as the foundation for subsequent harmonization, auditing, and quantitative quality control analyses. This approach ensures that immunoglobulin format annotations are not only internally consistent but also aligned with community-accepted standards.

Controlled term	Dataset value	Definition (IMGT-based)	Required criteria	Exclusion criteria
H2L2 antibody	Antibody	A full-length immunoglobulin composed of two heavy chains and two light chains.	Two heavy and two light chains forming a canonical IgG-like architecture.	Fragment-only structures (e.g. Fab, scFv) are excluded.
scFv	scFv	A single-chain variable fragment arranged as VH–linker–VL or VL–linker–VH, consisting of one heavy-chain variable region and one light-chain variable region.	Exactly one VH and one VL connected by a covalent linker; no constant domains.	Constructs lacking a linker or containing constant domains are excluded.
VHH / nanobody	Nanobody/VHH	A single-domain antibody derived from camelids, consisting of a single variable domain homologous to the heavy-chain variable region.	Single variable domain; camelid origin; no light-chain domains.	Strictly restricted to camelid-derived VHH; non-camelid VH-only constructs are excluded.
BJ (light-chain dimer)	BJ	A Bence Jones-type or engineered light-chain dimer composed exclusively of light-chain domains forming a homodimer.	Absence of heavy-chain domains; two light-chain domains forming a dimer.	Includes Bence Jones proteins and engineered light-chain dimers; distinct from VHH.

Table 1. **Controlled immunoglobulin format vocabulary.** *Entries are classified using the controlled immunoglobulin format vocabulary, with categories including antibodies, scFv, VHH, and BJ.*

Audit Summary and Relabelling

Following controlled vocabulary harmonization, we conducted a systematic cross-source consistency audit to evaluate the correctness of immunoglobulin format annotations inherited from source databases. This audit compared reported molecular formats against controlled vocabulary criteria, using chain composition and species information as primary decision factors.

Nanobody (VHH) entries were prioritized for detailed inspection, as this category is particularly prone to mislabelling. VHH are strictly defined as antigen-binding domains derived from camelid heavy-chain-only immunoglobulins; however, several source records labelled as “nanobody” did not meet these criteria.

As a result of this audit, 217 entries originally labelled as VHH were relabelled. Of these, 96 entries were reassigned to BJ due to the absence of heavy-chain domains, and 121 entries were reassigned to scFv, because they exhibited VH-VL architectures or non-camelid species origins. All relabelling decisions were applied consistently across the dataset and recorded in the provenance field.

Distribution by Format and Species

After completion of relabelling, we quantified the composition of the ANDD dataset by

immunoglobulin format and species to assess the biological plausibility of the final annotations.

The post-QC format distribution shows that the dataset is composed of 18,464 antibody entries, and 30,336 non-antibody entries, of which 30,119 are VHH/nanobody, 121 are scFv, and 96 are BJ/light-chain dimers. ANDD is dominated by VHH entries (30,119 records) and antibody entries (18,464 records), reflecting the extensive use of these formats in therapeutic and experimental antibody research. Smaller numbers of scFv (121 records) and BJ (96 records) are also present, consistent with their more specialized experimental usage. These results are summarized in Fig. 2.

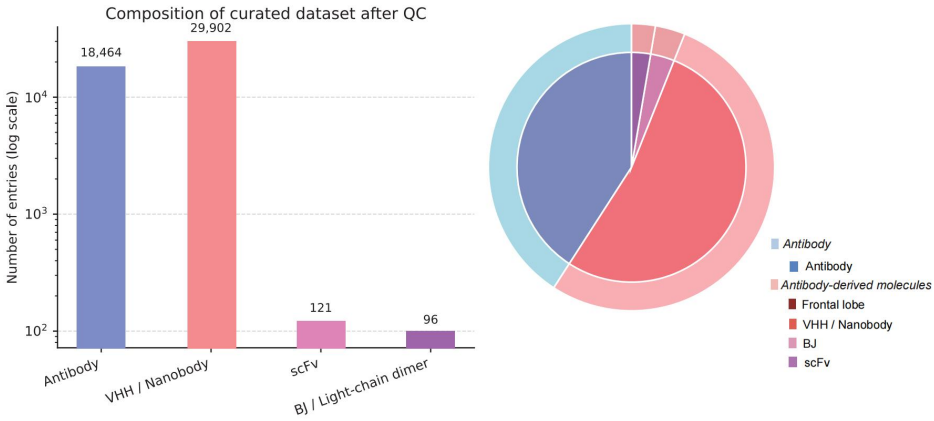

Fig. 2 **The quality control of entries' format.** *The left panel shows the composition of ANDD after quality control. The right panel displays a nested pie chart, where the outer ring distinguishes antibody (18,464) and antibody-derived molecules (30,336), and the inner ring further resolves antibody-derived molecules into VHH/nanobody, scFv, and BJ. Together, these panels provide a quantitative overview of ANDD after quality control.*

Species-level analysis further reflects the characteristics of ANDD data distribution. The dataset exhibits a pronounced long-tailed distribution, indicating that most entries are concentrated in a limited number of well-studied species. Homo sapiens (8,290 entries, 38.3%) and Mus musculus (7,074 entries, 32.7%) constitute the largest sources, together accounting for the majority of records, Camelid-related species form the next most prominent group. The top ten species collectively cover most records, while numerous additional species contribute only small numbers of entries, reflecting a typical long-tail pattern. When aggregated by biological category, mammalian sources overwhelmingly predominate, whereas other categories represent only minor fractions, supporting the biological consistency of the ANDD. As shown in Figure 3, VHH annotations are exclusively associated with camelid species, including Camelidae, Lama glama, and Vicugna pacos, consistent with their biological origin. In contrast, antibodies, scFv, and BJ formats predominantly originate from Homo sapiens and Mus musculus, reflecting common sources for antibody engineering and discovery. The observed species–format relationships are biologically coherent and further indicate the accuracy of our dataset.

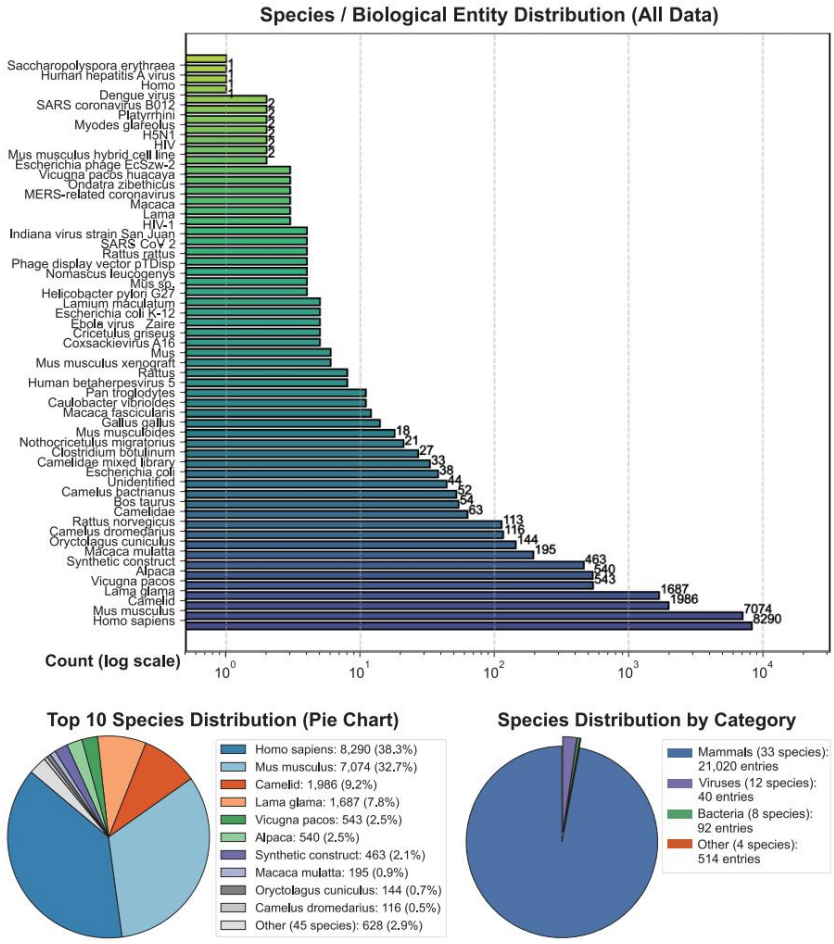

Fig. 3 **The quality control of entries' species**. *Species-level annotation revealed broad taxonomic coverage, with the majority of entries derived from Homo sapiens (8,290 records) and Mus musculus (7,074 records), followed by camelid species including Lama glama (1,687), Vicugna pacos (543), Camelus dromedarius (116), and Camelus bactrianus (52). Additional records originate from rabbit, macaque, bovine, rodent, microbial, viral, and synthetic construct sources.*

Status Flagging and Ambiguity Resolution

Each ANDD record includes a provenance field including the status flag indicating whether the original immunoglobulin format annotation was retained or modified during quality control.

The provenance allow users to distinguish between relabelled entries and original entries, facilitating informed downstream analyses. Importantly, after completion of the audit, no unresolved ambiguities remained. All records could be confidently assigned to a single immunoglobulin format based on controlled vocabulary criteria. Entries lacking sufficient evidence to support their original annotation were either corrected or excluded during QC, ensuring that the released dataset adheres to clearly defined annotation standards, guaranteeing the accuracy and consistency of ANDD.

We also distilled a concise QC checklist that captures the key quality assurance measures applied to the ANDD dataset. This checklist reflects the establishment of an IMGT-based controlled vocabulary for immunoglobulin formats, systematic cross-source consistency checks,

explicit documentation and quantification of relabelled entries, and comprehensive post-QC summaries by molecule format and species, shown in Table 2.

QC Item	Description	Method / Criteria	Outcome	Reference
Controlled vocabulary	Standardized immunoglobulin format terminology	IMGT used as standard authority; four mutually exclusive formats defined (Antibody, scFv, VHH, BJ)	Completed	Table 2
Formal definitions	Explicit definitions and decision rules provided	Definition, required criteria, and exclusion criteria specified for each format	Completed	Table 2
Cross-source check	Verification of format annotations across sources	Chain composition and species origin compared against controlled vocabulary	Completed	QC Report
Relabelled entries	Corrections to misannotated records identified and counted	217 VHH entries relabelled (96 → BJ, 121 → scFv)	Completed	QC Report
Relabelling rationale	Biological rationale for annotation changes recorded	Non-camelid origin or incompatible chain architecture	Completed	QC Report
Distribution (Format)	Post-QC counts summarized by immunoglobulin format	Antibody: 18,464; VHH: 30,110; scFv: 121; BJ: 96	Completed	Figure 8
Distribution (Species)	Post-QC counts summarized by species	Human, mouse, and camelid species distributions quantified	Completed	Figure 9
Consistency check	Biological plausibility of format–species associations	VHH restricted to camelid species; others predominantly human/mouse	Completed	QC Report
Unresolved ambiguities	Records evaluated for remaining annotation uncertainty	No unresolved ambiguities identified	Completed	QC Report
Status flag	Annotation change tracking enabled	Status flag included in provenance field per record	Completed	QC Report
QC transparency	All QC decisions traceable in released dataset	Provenance records retained for all modified entries	Completed	Metadata

Table 2. **QC check list.**